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# Palaeomagnetism of the Särna alkaline body

R. L. SMITH and J. D. A. PIPER

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**Additional palaeomagnetic results for the Särna alkaline body of western Sweden confirm the study of Bylund & Patchett (Bylund, G. & Patchett, P. G., 1977: Palaeomagnetism and Rb–Sr isotopic evidence for the age of the Särna alkaline complex, western central Sweden. *Lithos 10*, 73–79). A combined mean direction ($D = 201°$, $I = -23°$) from both studies yields a palaeomagnetic pole at 13.6° W, 38.1° S ($dp$ 3.9°, $dm$ 7.3°) correlating with other European Upper Carboniferous data. Thermomagnetic determinations and petrologic study suggests that pyrrhotite is the predominant mineral responsible for magnetic remanence.** □ *Palaeomagnetism, thermomagnetism, pole positions, Särna alkaline body, petrography, pyrrhotite, syenite, Upper Carboniferous, Särna, north-western Dalarna, west-central Sweden, N6142 N6142 E1254 E1254.*

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The Särna alkaline body is a poorly-exposed, cancrinite-nepheline syenite intruding Proterozoic sediments over 9 km² near Särna in western Sweden (12.9° E, 61.7° N). Palaeomagnetic and radiometric data for this body have been given by Bylund & Patchett (1977). Here we note additional palaeomagnetic results which confirm the work of these authors and help to define the mean palaeomagnetic pole for the body.

Our samples were drilled from sporadic outcrop near the summit of the hill Ekorråsen and oriented by sights on the sun. Thirteen samples exhibited stable and systematic behaviour to progressive alternating field (a.f.) treatment. All these samples (initial natural remanent magnetisations (NRM's) in the range $0.02 \rightarrow 1.47 \times 10^{-5}$ A·m²/kg) showed moderate fall in moment with progressive treatment to 140 mT (milliTesla) for higher coercivity samples (Fig. 1) although low coercivity samples became too weak to measure after 20 mT. After treatment in low fields, all but four samples achieved shallow negative inclinations and southerly declinations (Fig. 2). Four samples with other magnetic properties typical of the remainder of the collection stabilise with moderate to steep positive inclinations; we have no concise explanation for these four directions but Bylund & Patchett (1977) measured one normally magnetised sample and it is possible that these samples record transitional field directions; they are excluded from the group mean calculation (Table 1).

The a.f. cleaned directions of this study are plotted in Fig. 2 together with the data of Bylund & Patchett; the two sets of data are very comparable and the combined mean for all 19 samples is listed in Table 1. As noted previously, the palaeomagnetic pole correlates with other European Late Carboniferous data and is consistent with the $287 \pm 14$ Ma isochron of Bylund & Patchett.

*Table 1.* Summary of palaeomagnetic results from the Särna alkaline body.

$D$ and $I$ are the mean declination and inclination of the magnetisation derived from $N$ samples with precision parameter $k$ and 95% radius of confidence, $\alpha_{95}$, about the mean direction; $dp$ and $dm$ are the radii of the oval of 95% confidence about the palaeomagnetic pole in the colatitude direction and at right angles to it respectively.

| | $D$ | $I$ | $N$ | $k$ | $\alpha_{95}$ |
|---|---|---|---|---|---|
| This study | 202.3 | −24.4 | 8 | 40.7 | 8.8 |
| Bylund & Patchett | 200 | −22 | 11 | 18.0 | 11.0 |
| Combined mean | 201.1 | −23.2 | 19 | 25.0 | 6.9 |

Palaeomagnetic pole: 13.6° W, 38.1° S ($dp = 3.9°$, $dm = 7.3°$)

Thermomagnetic determinations were made on small size (~ 1 mm) samples using a Chevallier type balance. Saturation magnetisation ($J_S$) *versus* temperature curves are not typical of titanomagnetite-held remanence (Fig. 3) and all exhibited a characteristic fall in $J_S$ with temperature; no single Curie point was identified in any sample except No. 2 illustrated in Fig. 3 with a possible Curie point at 580°C close to that of

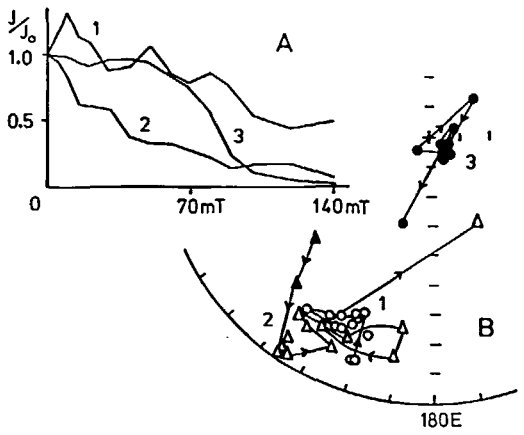



*Fig. 1.* □A. Changes in normalised intensity expressed as a function of initial remanence, $J_o$, during progressive alternating field demagnetisation (field values in mT). □ B. Changes in direction of magnetisation; equal area projection – upper and lower hemisphere plots represented by open and closed symbols respectively. (For clarity only a few representative directions are shown.)

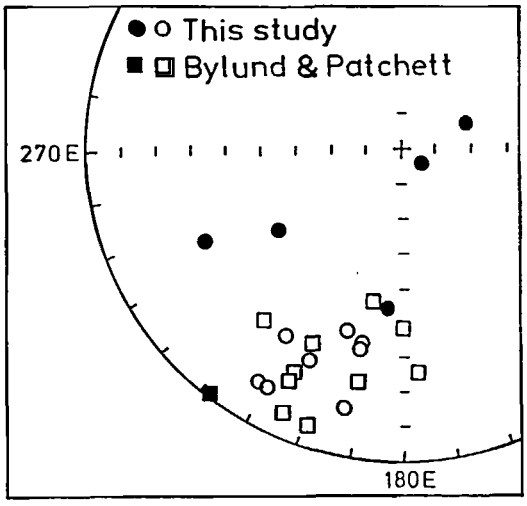



*Fig. 2.* Comparison of a.f.-cleaned, magnetic-vector directions from this study with the data of Bylund & Patchett (1977). Open symbols denote upper-hemisphere plots, filled symbols denote lower-hemisphere plots.

pure magnetite. Heating and cooling curves are similar or identical (Fig. 3) showing that little or no chemical change has been promoted by heating. A polished-section study revealed only disseminated granular sulphides less than 10 $\mu$m in size; these were too fine to be positively

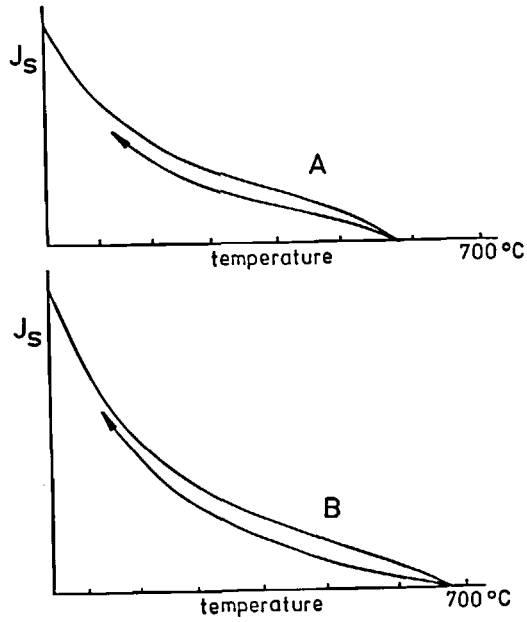



*Fig. 3.* Change in saturation magnetisation ($J_S$) with temperature. $J_S$ axis is arbitrary. □ A. Sample No. 2. □ B. Sample No. 4.

identified although a pale cream colour and bluish tinge suggested that some pyrrhotite was present. The presence of pyrrhotite as a magnetic-remanence carrier would be consistent with the saturation magnetisation behaviour illustrated in Fig. 3 with only weak ferrimagnetic properties being present above 300°C (Nagata 1961; Bhimasanharam 1964), although little appears to be known about the magnetic stability of this mineral.

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